

3(a)(i)	any two from: spherical / ball shaped ; (water-) soluble ; amino acids with hydrophilic R groups / hydrophilic amino acids, on outside of molecule ; A amino acids with hydrophobic R groups in centre of molecule physiological / metabolic, role ; carbonic anhydrase is an enzyme, (enzymes are globular proteins) ;	2
3(a)(ii)	any two from: catalyses / AW, reaction between water and carbon dioxide ; (to form) carbonic acid ; catalyses formation of carbon dioxide and water from carbonic acid in lungs ;	2
3(b)(i)	<u>guanine</u> ; any two from: double ring (base) / purine (base) ; forms three hydrogen bonds with complementary base ; AVP ; e.g. detail of difference between adenine and guanine	3

Question	Answer	Marks
3(b)(ii)	any four from: three bases / triplet of bases / codon, code(s) for one amino acid ; frameshift mutation ; change in, triplet of bases / codon ; <i>ref. to every</i>, triplet / codon, after deletion will be different ; may introduce a stop codon ; primary structure / sequence of amino acids / AW, will be altered ; (stop codon results in) formation of, shortened / truncated, polypeptide ; AVP ; e.g. <i>ref. to</i> errors in RNA splicing <i>ref. to</i> affects post-translational modification bonds that form between R groups, will / may, be altered tertiary structure of protein will be altered non-functional protein formed altered function of the protein	4
3(b)(iii)	any one valid suggestion e.g.: no effect on protein structure (because introns are non-coding) ; non-functional / reduced function of, protein (as intron not spliced after transcription) ; may affect regulation of gene expression ; A example e.g. reduction in the rate of transcription RNA polymerase does not bind <i>ref. to</i> errors in RNA splicing ; <i>ref. to</i> formation of an oncogene ;	1